INSTITUTO FEDERAL
Amazonas
Campus Manaus Centro

1ª Olimpíada de Derivadas
Diretoria de Pesquisa e Extensão
Centro Acadêmico de Licenciatura em Matemática - CALMAT

IFAMdx

Derivando no CMC

Sugestões de questões

Lista elaborada a partir de questões de olimpíadas e materiais de instituições públicas e privadas, como UTFPR, UnB, UFSC, Unicamp, PUC-PR e Univali, incluindo itens adaptados e itens reproduzidos com fins educacionais.

Modalidade Júnior

Tópicos: Funções elementares e regra da cadeia

1. (UTFdx) $y = x(x^2)(x^3)(x^4)...(x^{2025})(x^{2026})$
2. $f(x) = x^{-1/3} + 3x^5 + 2x^{-3} + 5x^{-1}$
3. $f(x) = 15x^{100} - 3x^{12} + 5x - 46$
4. $y = 9x^{1001} + 4x^{-2/3} + \pi x^7$
5. $y = x^8 + (2x+4)^3 + \sqrt{x}$
6. $y = \dfrac{x+1}{\sqrt{x^2-3}}$
7. $y = \sqrt[3]{6x^2+7x+2}$
8. $f(x) = \sqrt[5]{x^4} + 2x^3 - \dfrac{1}{5}x^{5/3}$
9. $f(w) = aw^2 + b$
10. $f(x) = (2x+1)(3x^2+6)$
11. $f(x) = (x-1)(x+1)$
12. $f(s) = (s^2-1)(3s-1)(5s^3+2s)$

13. $f(x) = 7(ax^2 + bx + c)$

14. $y = (x^4 + 14)^{3x} - 3x^2 + 2x - 7$

15. $f(x) = \dfrac{2x+4}{3x-1}$

16. $f(x) = \dfrac{x+1}{x+2}\,(3x^2 + 6x)$

17. $f(t) = \dfrac{1}{2}\,(t^2 + 5)\,(t^6 + 4t)$

18. $f(t) = \dfrac{(t-a)^2}{t-b}$

19. $y = 2x(2 + 3x)^5 + \sqrt{2 + x}$

20. $y = (x - 1)\sqrt{x}$

21. $y = x\sqrt{x^2 - 3x + 2}$

22. $y = (3x^5 - 1)^7$

23. $y = x^2\sqrt{3} + 5\sqrt{x^3} + \sqrt{5}$

24. $f(x) = 3e^x + 10x^3 \ln x$

25. $y = \sqrt{3x^2 - 1} - \dfrac{1}{2}\sqrt{(1 - 2x)^3}$

26. $g(x) = \ln(x^{-4} + x^4)$

27. $f(x) = \dfrac{2x^4 - 3}{x^2 - 5x + 3}$

28. $y = \dfrac{2}{\sqrt{x}} - \dfrac{\ln(x^2 - 1)}{\sqrt[4]{x^7}} - 1$

29. $y = \dfrac{5e^x}{3e^x + 1}$

30. $f(x) = \dfrac{7^x 3^x}{5^{2x}}$

31. $f(x) = \dfrac{1}{\ln x} + \dfrac{1}{(\ln x)^2} + \dfrac{1}{(\ln x)^3}$

32. $y = \dfrac{2^x 3^x}{5^x}$

33. $f(x) = (x^x)^x$

34. $y = \sqrt{x + \sqrt{x + \sqrt{x}}}$

35. $y = (x^2 + 5x + 2)^7$

36. $y = \left(\dfrac{3x+2}{2x+1}\right)^5$

37. $y = (3x^2+1)^3(x-x^2)^2$

38. $f(x) = 5\sqrt{x^2+3}$

39. $y = 3^{2x^2+3x-1}$

40. $y = \log_2(3x^2+7x-1)$

41. $y = \ln\left(\dfrac{e^x}{x+1}\right)$

42. $y = (x^2+1)^{2x-1}$

43. $y = \ln^3\sqrt{5x+\sqrt{x}}$

44. (UTFdx) $y = \log_2(\log_3(\sqrt[5]{x^\pi \log_4 x}))$

45. $y = \dfrac{1}{x} - \ln(e^x) + e^{-x}$

46. $y = x^x$

47. (UTFPR Cornélio) $y = (x^2+2)^{10}.e^{\ln(3x)}$

48. (UNIVALI Itajaí) $y = e^{1/x}.\sqrt{(1+3x^3)^3}$

49. (PUCPR) $y = \dfrac{\sqrt{x+1}-\sqrt{x-1}}{\sqrt{x+1}+\sqrt{x-1}}$

50. $f(x) = e^{(x-1)/(x+1)}$

51. $f(x) = \sqrt{\dfrac{\ln x}{e^x}}$

52. $f(x) = \sqrt{\dfrac{x+1}{3x-2}}$

53. $f(x) = \dfrac{x^3+1}{x+1}, x \neq -1$

54. $y = x^x.e^{x^2}.\ln x.\sqrt{x+1}$

55. $y = x^2.\ln x.e^x$

56. (JEE) $y = \dfrac{x^{1/2}(1-2x)^{2/3}}{(2-3x)^{3/4}(3-4x)^{4/5}}$

57. $y = (1+1/x)^{x^2}$

58. $y = \dfrac{\sqrt{x-1}}{\sqrt[3]{(x+2)^2}\sqrt{(x+3)^3}}$

59. $y = (\log x)^x + x^{\log x}$

60. $y = x^{x^2+1}(1 + \ln x^2)$

61. $y = \dfrac{x^2}{2} + \dfrac{1}{2}x\sqrt{x^2+1} + \ln\sqrt{x + \sqrt{x^2+1}}$

62. $y = \sqrt[3]{\dfrac{x(x^2+1)}{(x-4)^2}}$

63. $f(x) = \dfrac{(x-1)^2}{(x-2)(x+3)}$

Modalidade Sênior

Tópicos: Funções elementares, regra da cadeia, trigonométricas e trigonométricas inversas, derivação implícita, derivada de ordem superior

1. (UTFdx) $y = x^{x^x}$

2. $y = 5^{\sin x}$

3. $y = \dfrac{1}{3}\sin(\cos(x^4)) - 2\tan(2x)$

4. $y = \sec\left(\dfrac{2x}{1+x^2}\right)$

5. $f(x) = \sin(3x^2 + x)$

6. $y = \dfrac{e^{(x\cos x)}}{x^2\cos x}$

7. $y = \ln(cos^2 x) - \dfrac{2}{3}e^x\ln(x^3)$

8. $f(x) = x^{\cos(x^x)}$

9. $y = \dfrac{\sin x + \cos x}{\tan x + \sec x}$

10. $f(x) = \dfrac{\sec^2(1/x)}{\tan^2(1/x)}$

11. $\sqrt{x+y} + \sqrt{xy} = \pi^2$

12. $f(x) = x^{\sin(x)\cos(x)}$

13. $y^2 + xy^x = 1 + y^{x^2}$

14. $y = \left(\cos(3x)e^{\cos(3x)}\right)^{\cos^3(3x)}$

15. $y = \ln\left(\sqrt{x} + \sin(e^{\sqrt{x}})\right)$

16. $y = \csc^2\left(\ln\left(\dfrac{e^x\tan^2(x)}{x^2+1}\right)\right)$

17. $y = \ln\left[(e^{2x^4+x^2}).(3x^2+2)\right]$

18. $y = (x^2+2)^x$

19. $y = \left[\sin^2(\sin(x+1)) + \cos^2(\sin(x+1))\right]^{x^2}$

20. $y = (x\ln x)^9$

21. $y = 7^{\ln\left(\frac{2}{x}\right)}$

22. $y = \sin^2(\cos(x^2)+1)$

23. $y = \ln(\ln(\ln(x)))$

24. $y = \sin(\sqrt{1-2^x})$

25. $y = \log_2\left(\sqrt{\frac{1+\sin^2(2x)}{\sqrt{1+x}}}\right)$

26. $y = e^{x^2}(x^3+2)^{10}\sqrt{x}$

27. $y = \cos(\ln(\sqrt[3]{\tan(x)}))$

28. $y = 2026^{(\cot x)}\left[\sec^2((2026x)^x) + (2026x^x) - \tan^2((2026x)^x)\right]$

29. $e^{xy} + e^x + e^y = x\cos(x+y)$

30. $y = \cos(x^{10})^{e^{\sin(\sqrt{x})}}$

31. $y = \sin(\cos(3x)).\sqrt{x.e^x}$

32. $y = \frac{x^{x^2}}{x^{\sqrt{x}}}$

33. $\frac{\ln(xy)}{xy^2} = 1$

34. $y = (cos^2(x^2) - \sin^2(x^2))(1-\sin^2(x^2))(2\cos^2(x)-1)$

35. $y = x^{\sin^2(x^2y+1)+\cos^2(x^2y+1)} + \tan^3(x.\sec(\sqrt{2x}))$

36. $y = y^x - x^y$

37. $y = \frac{(x^2+x+1)^{\cos^2(x)}.(x^2+x+1)^x}{(x^2+x+1)^{\sin^2(x)}.(x^2+x+1)^{\cos(x+x)}}$

38. $y = \frac{e^{\sin^2(x)}.e^{\tan^2 x}.e^{\sqrt{x}}.e^1}{e^{-\cos^2 x}.e^{\sec^2 x}.e^x}$

39. $y = \sqrt{\dfrac{\dfrac{1}{\frac{x}{x^2}} + \dfrac{1}{\frac{\sin(x)}{\cos(x)}} + \dfrac{1}{\frac{e^x}{e^{3x}}}}{\dfrac{x}{\frac{x^3}{x^3}} + \dfrac{1}{\frac{\cos(x)}{\sin(x)}} + \dfrac{e^x}{\frac{1}{e^x}}}}$

40. $y = \left(\left(\left(\left(\left(\left(\left(2^{\frac{3}{2}}\right)^{\frac{\sin x}{2}}\right)^{\frac{\cos x}{2}}\right)^{\frac{\tan x}{2}}\right)^{\frac{\cot x}{2}}\right)^{\frac{\sec x}{2}}\right)^{\frac{\csc x}{2}}\right)^{\frac{x}{2}}$

41. (UTFdx) $y = \tan^4(3^{\sin x} + \tan(e^x))$

42. (UTFdx) $y = \log_5\left(\dfrac{\tan(x^2)}{e^x}\right)$

43. (UTFdx) $y = \sin^7(x^6 + \sin^5(x^4 + \sin^3(x^2)))$

44. (UTFdx) $e^{xy} + e^x + e^y = x\sin(x+y)$

45. (UTFdx) $y = \left(\dfrac{\sin\left(\frac{1}{x}\right)}{\tan\left(\frac{1}{x}\right)}\right)^{\pi x}$

46. (UTFdx) $y^2 = xy$

47. Calcule a segunda derivada das funçoes dadas:

    a) $f(x) = x\cos x$
    b) $f(x) = \sqrt{3}(e^x + x^3)$
    c) $f(x) = \sin x \cos x$

48. Calcule a terceira derivada das funções:

    a) $f(x) = 6x^5 + 4x + 5$
    b) $y = x\ln x$
    c) $y = xe^{-2x}$